Exact Lyapunov Weights for Multichannel Defect Kernels

Abstract

We solve an exact retained-state weighting problem with a regular state and multiple center, continuous, orbit, and topology defect channels. The solution quantifies the trade between rare entry, forced return, and defect persistence. It shows precisely what physical conditional estimates a state-weighted Yang–Mills argument must supply.

1 Star defect kernel

Let state 0 be regular and $i = 1, \dots, n$ label defect channels. Consider the positive subkernel

$$Q(0, i) = a_i, \qquad Q(i, 0) = b_i, \qquad Q(i, i) = c_i,$$

with all other entries zero and subprobability rows.

Theorem 1. *Fix $q > \max_i c_i$ and suppose $b_i/(q - c_i) \geq 1$. Set*

$$V(0) = 1, \qquad V(i) = \frac{b_i}{q - c_i}.$$

Then

$$QV \leq qV \quad \Longleftrightarrow \quad \sum_{i=1}^{n} \frac{a_i b_i}{q - c_i} \leq q.$$

Every defect row is exactly saturated when $b_i > 0$.

Proof. For a defect state,

$$\frac{(QV)(i)}{V(i)} = \frac{b_i}{V(i)} + c_i = q.$$

At the regular state, $(QV)(0)/V(0) = \sum_i a_i V(i)$, yielding the stated equivalence. □

Thus persistence cannot be hidden in the weight: necessarily $c_i < q$. When $b_i = 1$ and $c_i = 0$, the minimum feasible ceiling is

$$q_{\min} = \sqrt{\sum_i a_i}.$$

2 Four-channel calibration

For

$$q = 0.002430277501841998$$

and four forced-return, zero-persistence channels, equal entry allocation requires

$$\sum_i a_i = q^2 = 5.90624873596 \cdot 10^{-6}, \qquad a_i = 1.47656218399 \cdot 10^{-6}.$$

The charged weight is $V_i = 411.47564393$. The free-energy costs are

$$-\log q^2 = 12.03949966, \qquad -\log a_i = 13.42579402.$$

The scalar row ceiling is nevertheless one because a defect can force the following bad transition.

3 Frontier

The theorem explains the mechanism but does not provide the physical rates a_i, b_i, c_i. The retained local representation supplies center and orbit labels, while a genuine topological sector is absent on the current simply connected local block. Uniform conditional entry, persistence, and topology estimates remain open.